

Vectors and Vector-Valued Functions

ANSWER KEY



Vector operations in 3D

- 1 Let $\mathbf{a} = \langle 2, -1, 3 \rangle$ and $\mathbf{b} = \langle 1, 4, -2 \rangle$. Find:

a $\mathbf{a} + \mathbf{b} = \langle 3, 3, 1 \rangle$ **b** $\mathbf{a} \cdot \mathbf{b} = 2 - 4 - 6 = -8$ **c** $|\mathbf{a}| = \sqrt{14}$

- 2 Find the cross product $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 1, 0, 2 \rangle$ and $\mathbf{b} = \langle 3, 1, -1 \rangle$.

$\mathbf{a} \times \mathbf{b} = \langle (0)(-1) - (2)(1), (2)(3) - (1)(-1), (1)(1) - (0)(3) \rangle = \langle -2, 7, 1 \rangle$.

Vector-valued functions: velocity and acceleration

- 3 A particle's position is $\mathbf{r}(t) = \langle t^2, 3t, t^3 \rangle$.

a Find the velocity vector $\mathbf{v}(t)$. $\mathbf{v}(t) = \langle 2t, 3, 3t^2 \rangle$.

b Find the acceleration vector $\mathbf{a}(t)$. $\mathbf{a}(t) = \langle 2, 0, 6t \rangle$.

c Find the speed $|\mathbf{v}(1)|$. $\mathbf{v}(1) = \langle 2, 3, 3 \rangle$, speed $= \sqrt{4 + 9 + 9} = \sqrt{22}$.